

Object Oriented Programming 2

Practice Programming Exercise

(Scenario Based)

Ques1:

Design a class **memberType**.

1. Data members of memberType are **name of a person** (string), **member ID**(integer), **number of books bought**(integer) , and **amount spent**(float).
2. Include the member functions to perform the various operations on the objects of memberType—for example,
 modify, set, and show a person's name.
 Similarly, update, modify, and show the number of books bought and the amount spent.
 - a. Add the appropriate constructors.
 - b. Write the definitions of the member functions of memberType.
 - c. Write a program to test various operations of your **class** memberType.

Ques2 : You are to develop a class that represents fractions and performs typical arithmetic operations with them.

1. Define the **Fraction** class with two data members namely **numerator** and **denominator** of type long .
The constructor has two parameters of type long: the first parameter (numerator) contains the default value 0, and the second parameter (denominator) contains the value 1.

Declare operator functions as methods for +,-,*,/.
The operator functions of the binary operators +, -, *, / and
The input / output operators <<, >> are to be declared as friend functions of the Fraction class.
2. Implement the constructor for the Fraction class to obtain a positive value for the denominator at all times.
If the denominator assumes a value of 0, issue an error message and terminate the program.

- Write a main function that calls all the operators in the Fraction class as a test application. Output both the operands and the results.

Example for Arithmetic calculation with fraction

Implement operator functions according to given formulas

Addition	$\frac{a}{b}$	+	$\frac{c}{d}$	=	$\frac{a*d + b*c}{b*d}$
Subtraction	$\frac{a}{b}$	-	$\frac{c}{d}$	=	$\frac{a*d - b*c}{b*d}$
Multiplication	$\frac{a}{b}$	*	$\frac{c}{d}$	=	$\frac{a * c}{b * d}$
Division	$\frac{a}{b}$	/	$\frac{c}{d}$	=	$\frac{a * d}{b * c}$

Ques3:

Declare class “**Calculator**” which contains an attribute ‘**number**’ for storing integer value. It contains constructor to initialize a variable with value 1.

Class also contains two methods ‘**Square**’ and ‘**AddDigits**’ which perform the following tasks.

Square: It will take an integer as input and displays the square of a number.

(Ex. if you enter 4 , square is 16)

AddDigits: it will ask user to enter number and then displays the sum of all digits of a number.

(Ex. if you enter 467, sum of digit is 4+6+7 =17)

Demonstrate the functioning of Calculator by creating an object in “**main**” method.
On execution your program should display the following menu.

1. Display Square (i.e enter 1 for display square)
 2. Display Sum of Digits(i.e enter 2 for display sum of digits)
- Enter your choice: -----

Create an object of ‘**Calculator**’ class and call the method based on menu choice.

Sample output is as follows:

Sample output 1	Sample output 2
1. Display Square 2. Display Sum of Digits Enter your choice 1 Enter any number 7 The Square of number is 49	1. Display Square 2. Display Sum of Digits Enter your choice 2 Enter any number 3745 Summation of digits in number =19